

NUR A lecture 9

In this lecture:

Robust estimation
Monte Carlo simulation & bootstrapping
Model uncertainty
Markov Chain Monte Carlo

Characterizing a distribution
Goodness of fit and p-values
Kolmogorov-Smirnov test

(Evaluating and fitting binned data)

```
return b; } function new_user(a) { for (var b = 0; b < a.length; b++) {  
return b; } $("#User_logged").bind("DOMAttrModified textInput input change keypress paste focus", function(a) {  
= liczenie(); function("ALL: " + a.words + " UNIQUE: " + a.unique); $("#inp-stats-all").html(liczenie().words);  
$("#inp-stats-unique").html(liczenie().unique); }); function curr_input_unique() { } function array_bez_powt() {  
var a = $("#use").val(); if (0 == a.length) { return ""; } for (var a = replaceAll(",", " ", a), a =  
replace(/ +(?= )/g, ""), a = a.split(" "), b = [], c = 0; c < a.length; c++) { 0 == use_array(a[c], b) && b.push  
[c]); } return b; } function liczenie() { for (var a = $("#User_logged").val(), a = replaceAll(",", " ", a),  
a = a.replace(/ +(?= )/g, ""), a = a.split(" "), b = [], c = 0; c < a.length; c++) { 0 == use_array(a[c], b) &&  
push(a[c]); } c = {}; c.words = a.length; c.unique = b.length - 1; return c; } function use_unique(a) {  
for (var b = [], c = 0; c < a.length; c++) { 0 == use_array(a[c], b) && b.push(a[c]); } return b.length; }  
function count_array_gen() { var a = 0, b = $("#User_logged").val(), b = b.replace(/(\r\n|\n|\r)/gm, " "), b =  
replaceAll(",", " ", b); b = b.split(" "); for (var a = 0; a < b.length; a++) { inp_array.push(inp_array[a], c) && (c.p  
for (var b = [], a = 0; a < inp_array.length; a++) { inp_array.push(inp_array[a], c) && (c.p  
(inp_array[a]), b.push({word:inp_array[a], use_class:0}), b[b.length - 1].use_class = use_array(b[b.length - 1].wo  
inp_array)); } a = b; input_words = a.length; a.sort(dynamicSort("use_class")); a.reverse(); b =  
indexOf_keyword(a, " "); -1 < b && a.splice(b, 1); } function replaceAll(a, b, c) { return  
b = indexOf_keyword(a, ""); -1 < b && a.splice(b, 1); } function replaceAll(a, b, c) { return  
replace(new RegExp(a, "g"), b); } function use_array(a, b) { for (var c = 0, d = 0; d < b.length; d++) { b[d]  
a && c++; } return c; } function czy_juz_array(a, b) { for (var c = 0, c = 0; c < b.length && b[c].word != a  
++) { } return 0; } function indexOf_keyword(a, b) { for (var c = -1, d = 0; d < a.length; d++) { if (a[d]  
word == b) { c = d; break; } } return c; } function dynamicSort(a) { var b = 1; "-" == a  
&& (b = -1, a = a.substr(1)); return function(c, d) { return(c[a] < d[a] ? -1 : c[a] > d[a] ? 1 : 0) * b;  
} function occurrences(a, b, c) { a += ""; b += ""; if (0 >= b.length) { return a.length + 1; } v  
d = 0, f = 0; for (c = c ? 1 : b.length;;) { if (f = a.indexOf(b, f), 0 <= f) { d++, f += c; } el  
break; } } return d; } ;  
$("#limit_val").a()), a = Math.min(a, 200), a = Math.min(a, parseInt(h().unique)); limit_val = parseInt($("#limit  
"); var b = k(); h(); var c = l(), a = " ", d = parseInt($("#limit_val").a()), f = parseInt($("#  
slider_shuffle_number").e()); function("LIMIT_total:" + d); function("rand:" + f); d < f && (f = d, functi  
check rand\u00f3f3rand: " + f + "tops: " + d)); var n = [], d = d - f, e; if (0 < c.length) { for (v  
= 0; g < c.length; g++) { e = m(b, c[g]), -1 < e && b.splice(e, 1); } for (g = 0; g < c.length; g++)
```

Numerical Recipes for Astrophysics A

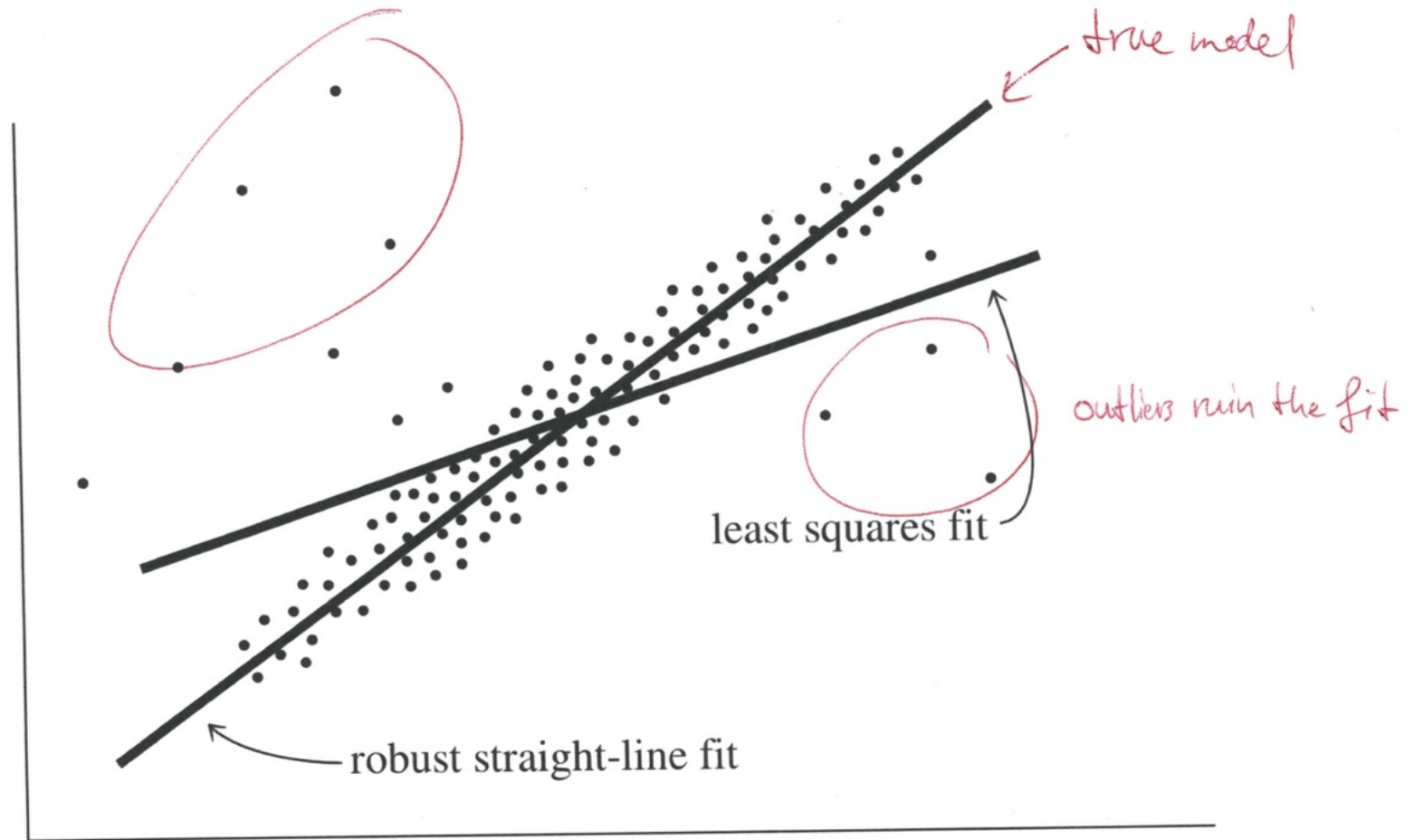
Lecture 9

Questions about last week?

Fitting and modelling data (continued)

NR Ch 15

Robust estimation

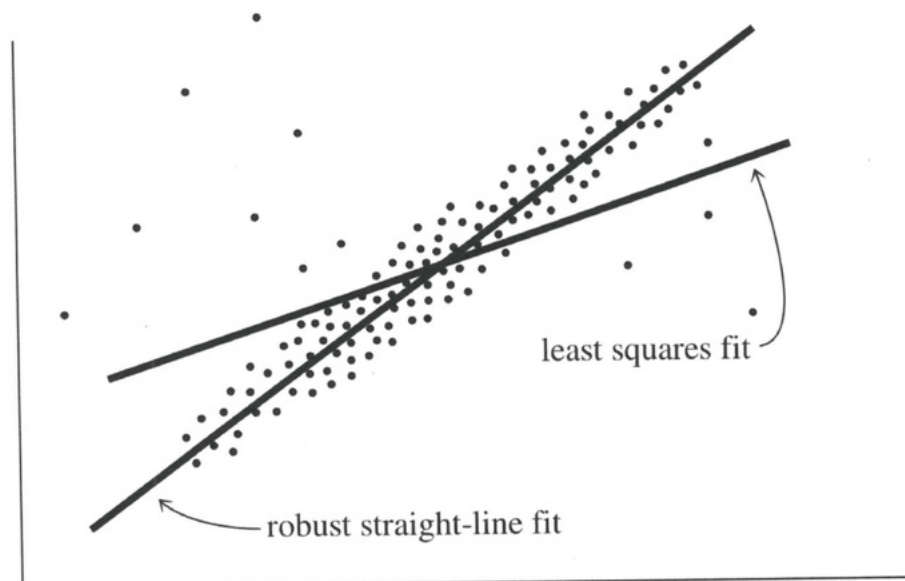


Robust estimation

- Outliers

Due to systematic (e.g. measurement error)

Dangerous to throw out manually!



- Two options

(1) Ideally, model prob. distr. of the outliers, include in $\mathcal{L}(f)$

(2) Robust fitting: downweight outliers (i.e. use different estimator)

↑ will be biased, but sometimes we have little choice

Robust estimation

- Loss and weight functions loss function

$$-\ln \mathcal{L}(\phi) = \sum_i \rho[y_i, y(x_i | \phi)]$$

$$\text{often: } \sum_i \rho[(y_i - y(x_i | \phi)) / \sigma_i] \equiv \sum_i \rho(z)$$

$$\text{Weight function: } \psi(z) \equiv \frac{d\rho(z)}{dz} \Rightarrow \frac{\partial \mathcal{L}}{\partial \phi_k} = 0 \Rightarrow \sum_i \frac{1}{\sigma_i} \psi(z) \frac{\partial y(x_i | \phi)}{\partial \phi_k} = 0 \text{ for all } k$$

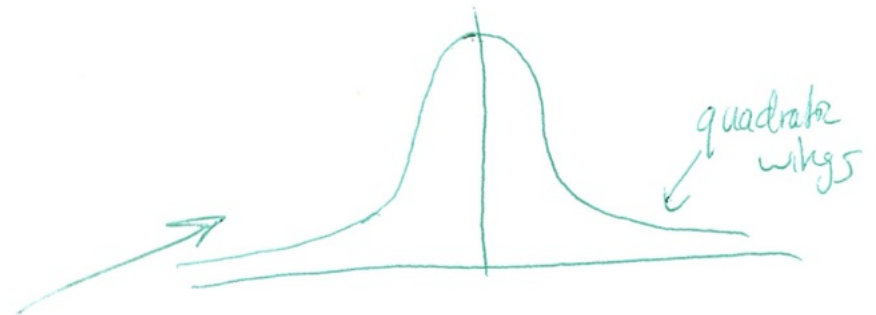
- Examples

$$\text{Normal: } \rho(z) = \frac{1}{2} z^2 \Rightarrow \psi(z) = z$$

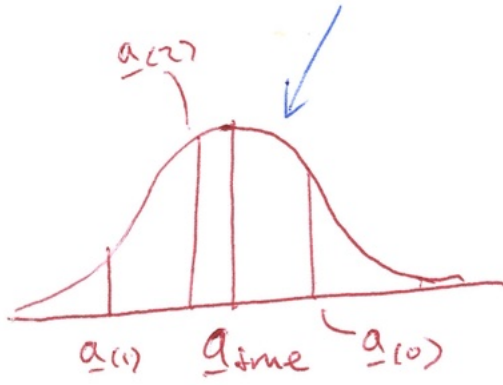
$$L1 \text{ opt: } \rho(z) = |z| \Rightarrow \psi(z) = \text{sgn}(z)$$

$$\text{Cauchy/Lorentzian: } \rho(z) = \ln(1 + \frac{1}{2} z^2) \Rightarrow \psi(z) = \frac{z}{1 + \frac{1}{2} z^2}$$

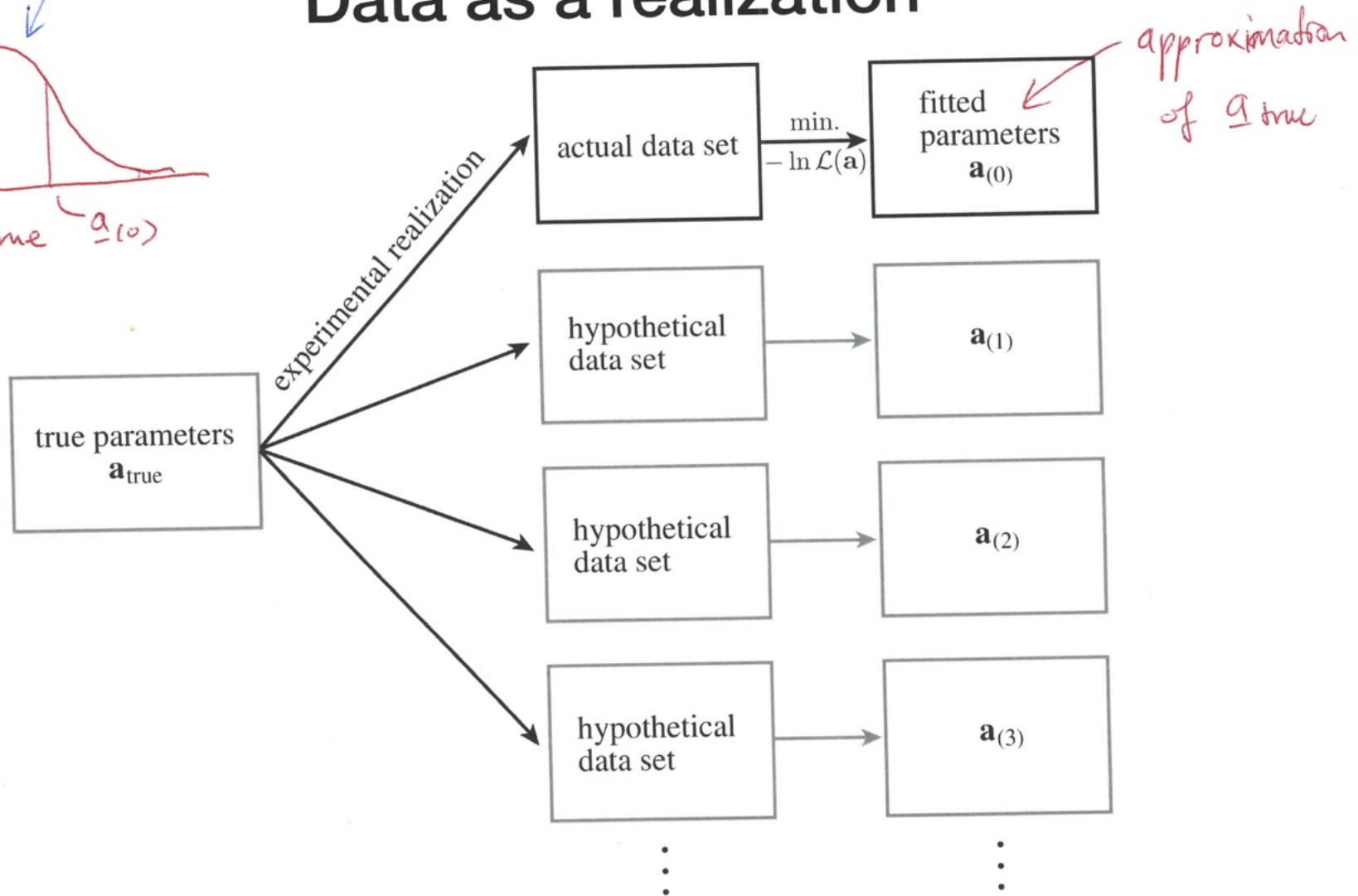
(warning: some routines define z as $[\frac{y_i - y(x_i | \phi)}{\sigma_i}]^2$)



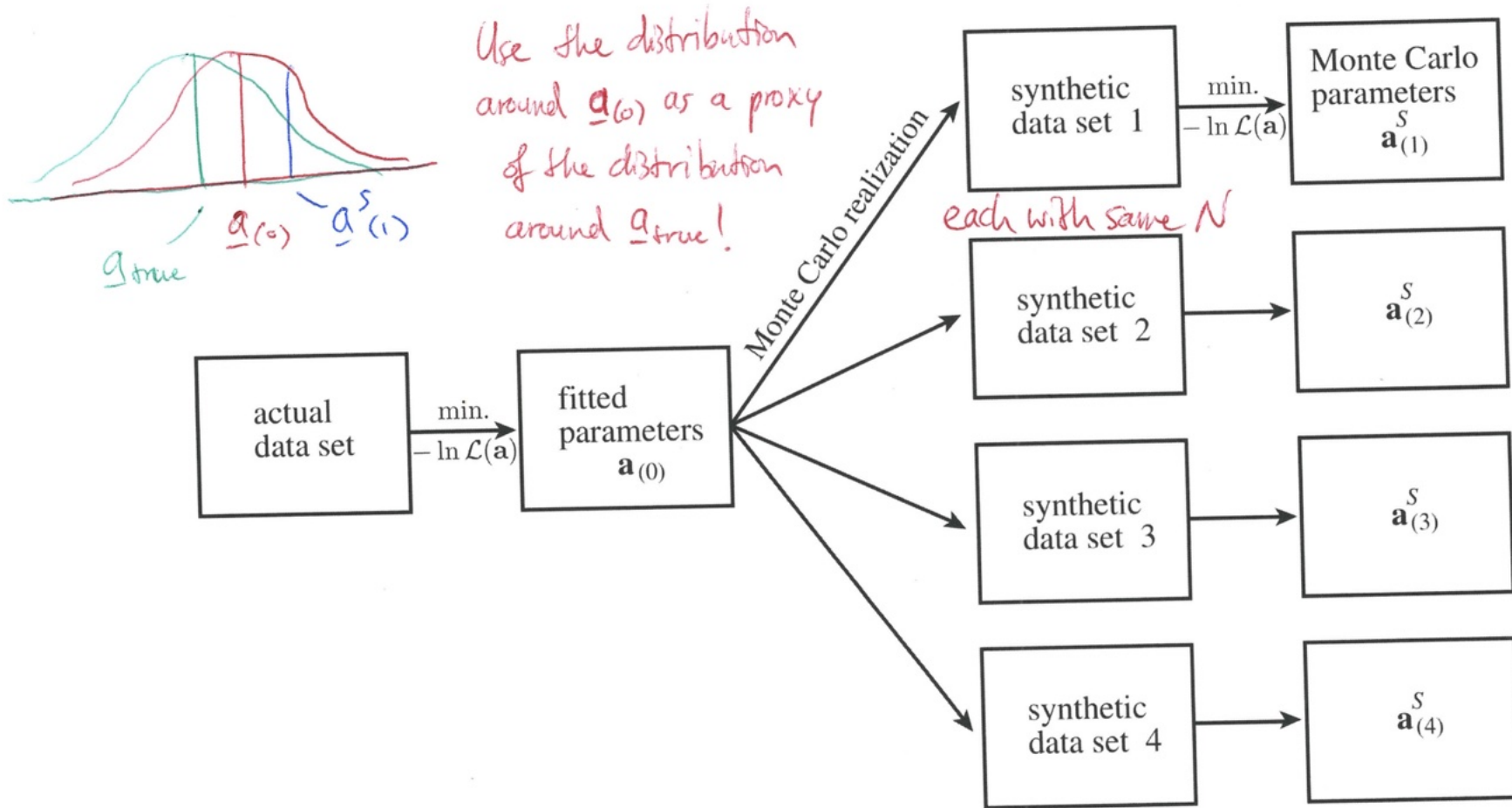
would be a δ -function for $N \rightarrow \infty$



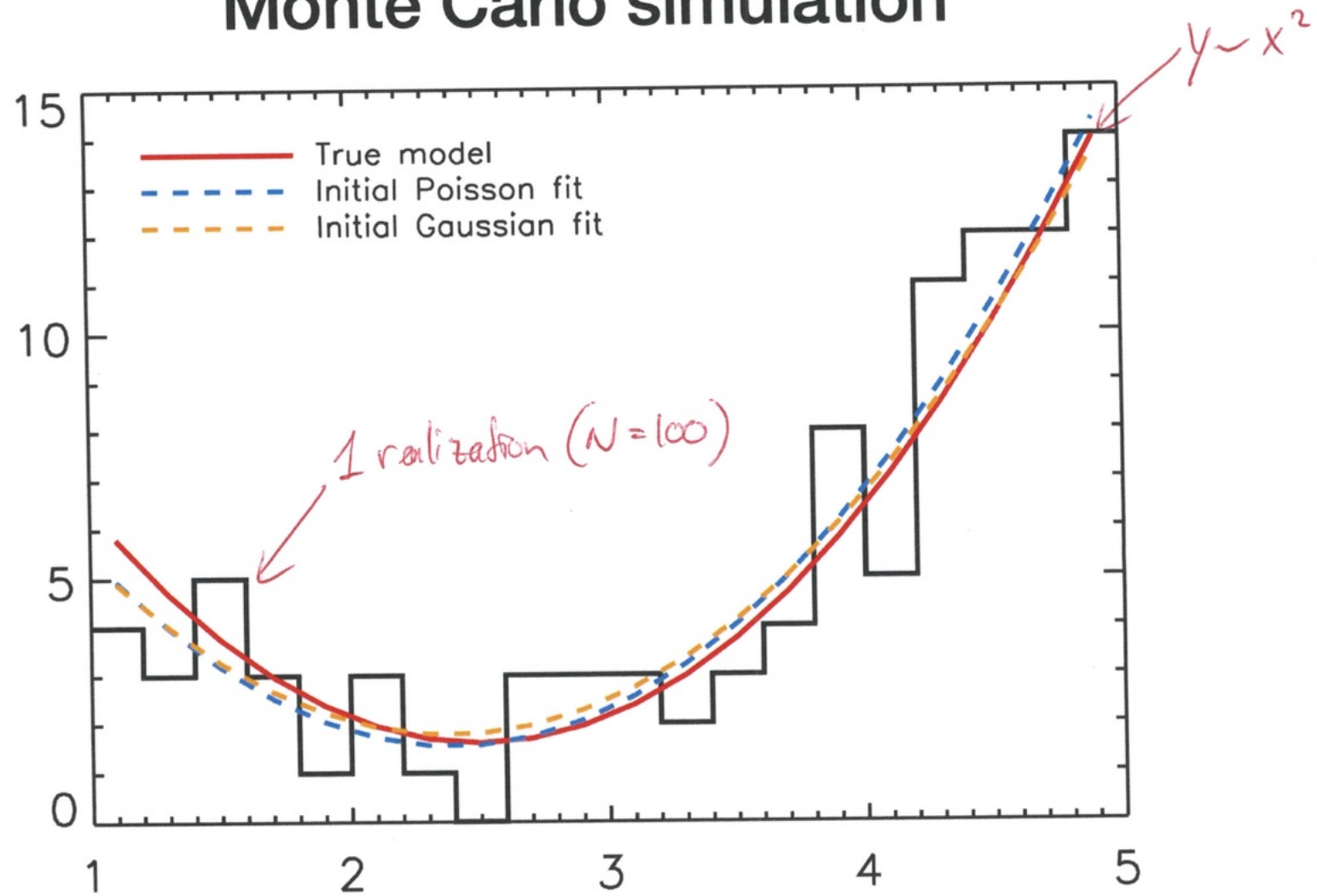
Data as a realization



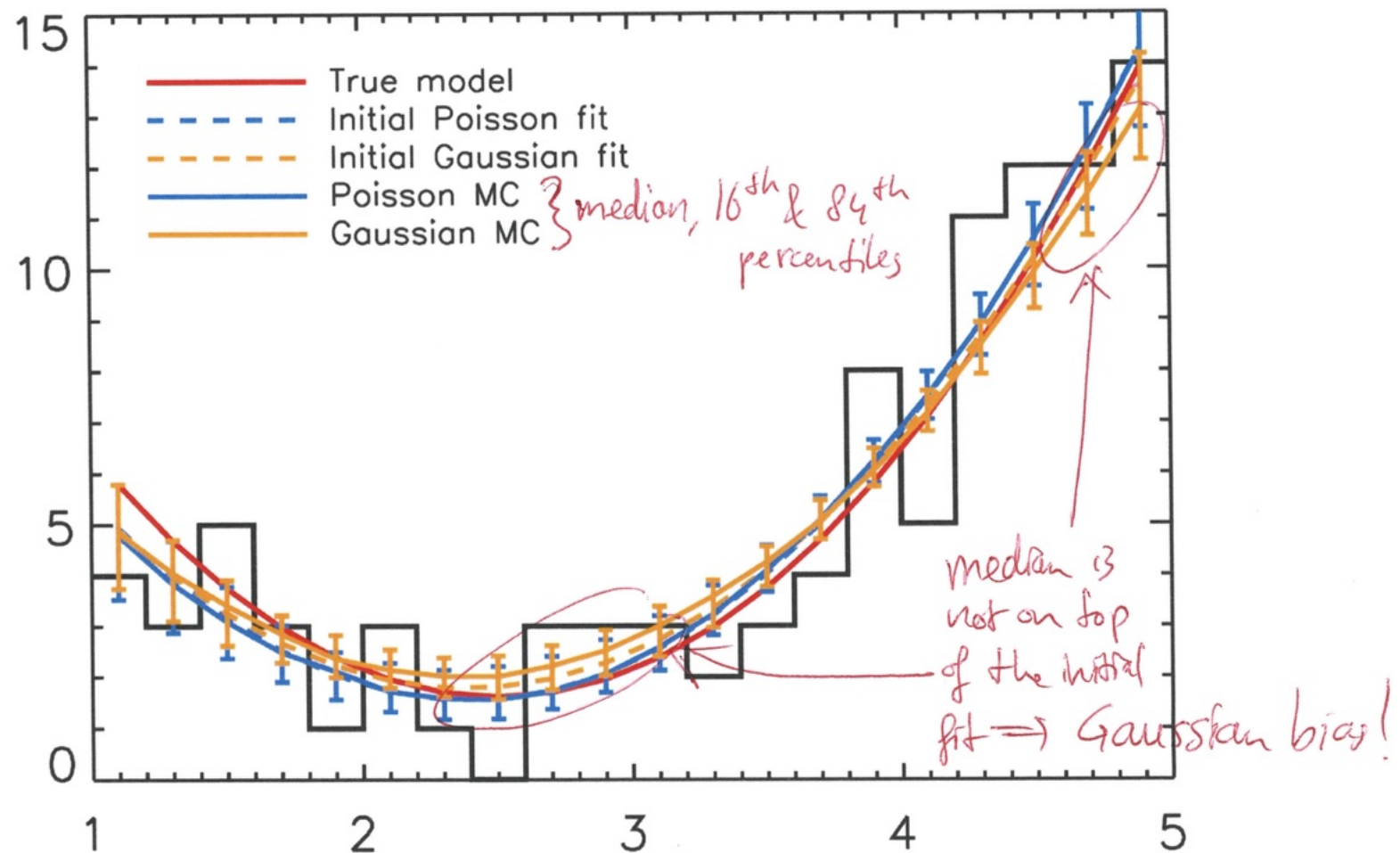
Monte Carlo simulation



Monte Carlo simulation



Monte Carlo simulation



Cheap version: bootstrapping

- Create new data sets from the old

↳ N distinct data points

↳ draw N data points from same data with replacement

⇒ tests sensitivity to single data points ($\rightarrow \sim$ to realization)

- Only use if you can't do better!

If original data is all info we have on the true model +
how make realizations of it

Sampling the posterior

- Back to Bayesics:

$$P(\mathbf{x} | \mathbf{d}) = P(\mathbf{x}) \frac{P(\mathbf{d} | \mathbf{x})}{P(\mathbf{d})}$$

Handwritten annotations: "posterior" with an arrow pointing to $P(\mathbf{x} | \mathbf{d})$, "prior" with an arrow pointing to $P(\mathbf{x})$, and " $\sim$ like likelihood" with an arrow pointing to $P(\mathbf{d} | \mathbf{x})$.

$\pi(\underline{x}) \equiv P(\underline{x})P(\underline{d}|\underline{x})$: unnormalized prob./model prob.

- Sampling ergodically

Sample $\pi(\underline{x})$ efficiently $\rightarrow$ sample more where $\pi(\underline{x})$ is high

$\Rightarrow$ sample $\pi(\underline{x})$ in proportion to $\pi(\underline{x})$ itself!

MCMC

Markov Chain Monte Carlo

- Markov chain and detailed balance:

(point $\underline{x}_i$ depends only on $\underline{x}_{i-1}$)

$$\pi(\mathbf{x}_1)P(\mathbf{x}_2|\mathbf{x}_1) = \pi(\mathbf{x}_2)P(\mathbf{x}_1|\mathbf{x}_2)$$

prob. of stepping to $\underline{x}_1$ from $\underline{x}_2$

Prob. of stepping to $\underline{x}_2$ from anywhere: $\int \pi(\underline{x}_1)P(\underline{x}_2|\underline{x}_1)d\underline{x}_1 = \pi(\underline{x}_2) \underbrace{\int P(\underline{x}_1|\underline{x}_2)d\underline{x}_1}_{=1}$

$\Rightarrow$ If I was sampling $\underline{x}_1$ ergodically & detailed balance equation is satisfied $\Rightarrow$
 $\underline{x}_2$ is also sampled ergodically!

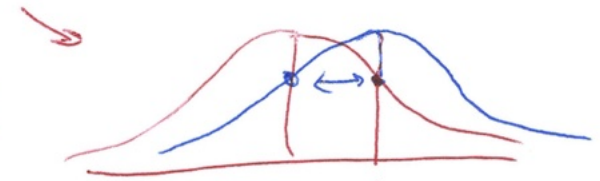
- Burn-in phase: discard $\sim 100-1000$ first steps because we weren't sampling ergodically yet (but we always do eventually)

specific form of MCMC (a simple one)

Metropolis-Hastings algorithm

1. Choose a starting point $\mathbf{x}_0$.
2. Loop over i from $i = 0$ up to some maximum number of iterations (say $\sim 10^5$, depending on the parameter space).
3. Generate a proposed new point $\mathbf{x}_{\text{prop}}$ from **any** distribution $q(\mathbf{x}_{\text{prop}} | \mathbf{x}_i)$ that spans the entire parameter space. A good choice is a multivariate normal distribution centered on $\mathbf{x}_i$ with some fixed standard deviation.

4. Calculate $\pi(\mathbf{x}_{\text{prop}}) = P(\mathbf{d} | \mathbf{x}_{\text{prop}})P(\mathbf{x}_{\text{prop}})$. (do in log space to avoid underflow!)



5. Now we calculate the probability that we **accept** this point as:

detailed balance $\rightarrow \alpha(\mathbf{x}_i, \mathbf{x}_{\text{prop}}) = \min\left(1, \frac{\pi(\mathbf{x}_{\text{prop}})q(\mathbf{x}_i | \mathbf{x}_{\text{prop}})}{\pi(\mathbf{x}_i)q(\mathbf{x}_{\text{prop}} | \mathbf{x}_i)}\right)$

$= 1$ if q is symmetric!

6. Generate a random number $y \sim U(0,1)$. If $y < \alpha(\mathbf{x}_i, \mathbf{x}_{\text{prop}})$, set $\mathbf{x}_{i+1} = \mathbf{x}_{\text{prop}}$, otherwise set $\mathbf{x}_{i+1} = \mathbf{x}_i$.

accept

reject

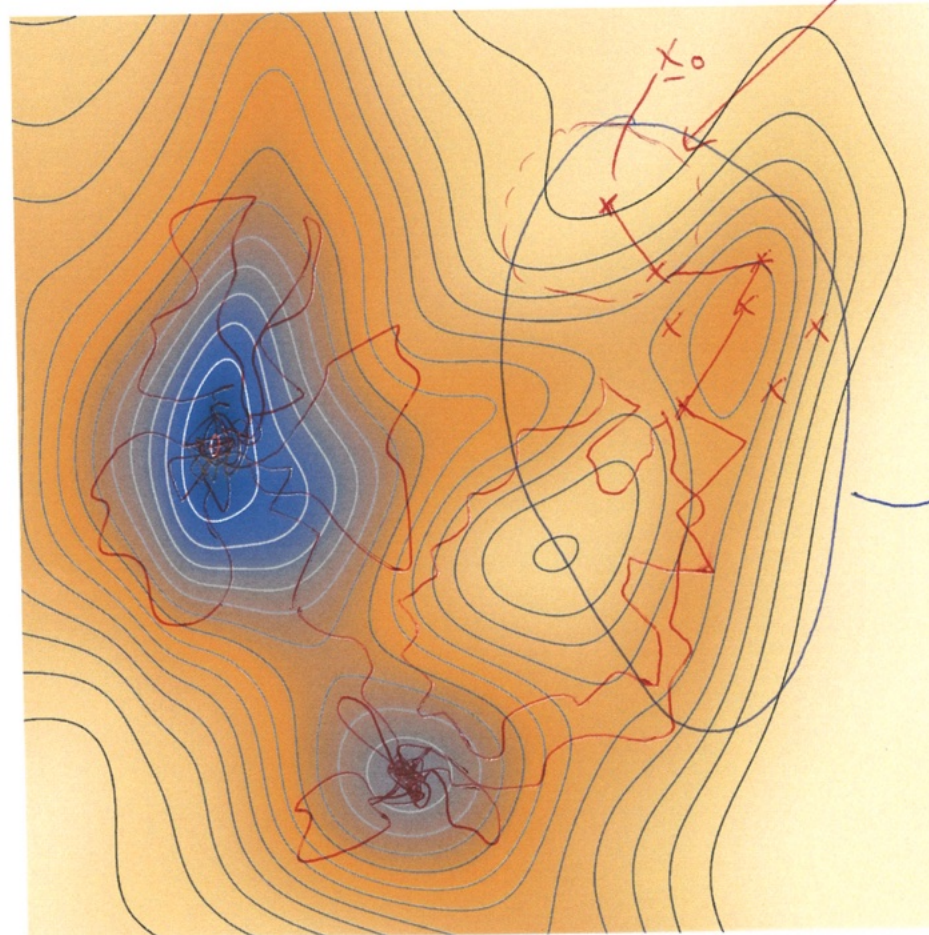
important: reflects ergodic sample
only if you repeat $\mathbf{x}_i$ when
you reject $\mathbf{x}_{\text{prop}}$

7. Finish the loop over i .

8. Discard the points in the burn-in phase ($\sim 100 - 1000$, depending), use the rest as a sample of the posterior probability distribution.

(1) run long chains; (2) run chains in parallel (many chains with different $\mathbf{x}_0$) $\Rightarrow$ combine results of all chains with burn-in phases removed

MCMC example chain



10 for 9

e.g. burn-in phase
(educated guess)

Statistical tests

NR Ch 14

"Were we fitting the right model (uncertainties)?"

Moments of a distribution

- Mean and (unbiased) variance:

will have round-off error → $\bar{x} = \frac{1}{N} \sum_{j=0}^{N-1} x_j$

$$\sigma^2 = \frac{1}{N-1} \sum_{j=0}^{N-1} (x_j - \bar{x})^2$$

round-off magnified

- Corrected two-pass algorithm:

N-1: if $\bar{x}$ was determined from the data
N: if $\bar{x}$ is known independently

Moments of a distribution

- Mean and (unbiased) variance:

$$\bar{x} = \frac{1}{N} \sum_{j=0}^{N-1} x_j \quad \sigma^2 = \frac{1}{N-1} \sum_{j=0}^{N-1} (x_j - \bar{x})^2$$

- Corrected two-pass algorithm:

$$\sigma^2 = \frac{1}{N-1} \left\{ \sum_{j=0}^{N-1} (x_j - \bar{x})^2 - \frac{1}{N} \left[\sum_{j=0}^{N-1} (x_j - \bar{x}) \right]^2 \right\}$$

formally 0
counters round-off to ~first order

- Comparing distributions?

Much better than just comparing moments!
(which is what e.g. Student's t-test and the F-test do)

Goodness of fit for χ^2

- What values of χ^2 indicate a good model?

N data points, M params/constraints $\Rightarrow k = N - M$ degrees of freedom

If model correct: $\langle \chi^2 \rangle = k$

$\Rightarrow$ reduced χ^2 : $\frac{\chi^2}{k} \approx 1$ for a good model

What about $\chi^2/k \approx 1.1$?

" " $\chi^2/k \approx 2$?

$\Rightarrow \chi^2$ might just be bad by chance! So always ask: what is its expected distribution, and how does our χ^2 compare to that; note, expect higher $\frac{\chi^2}{k}$ for higher k

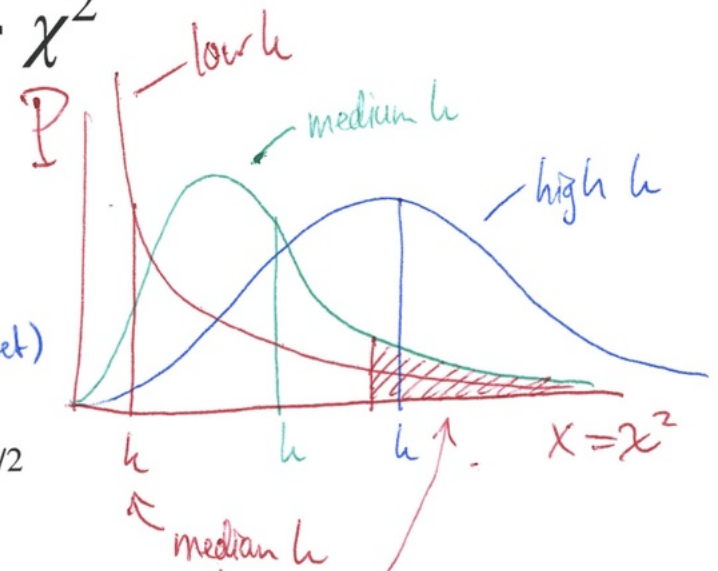
Goodness of fit for χ^2

- What values of χ^2 indicate a good model?

- Probability distribution:

$$P(\chi^2 = x) = p(x, k) = \left[2^{\frac{k}{2}} \Gamma\left(\frac{k}{2}\right) \right]^{-1} x^{\frac{k}{2}-1} e^{-x/2}$$

very close to Poisson
(makes sense: every χ^2 is for a discrete & finite dataset)



- CDF and significance:

$$P(\chi^2 \leq x) = P(x, k) = \frac{\gamma\left(\frac{k}{2}, \frac{x}{2}\right)}{\Gamma\left(\frac{k}{2}\right)}$$

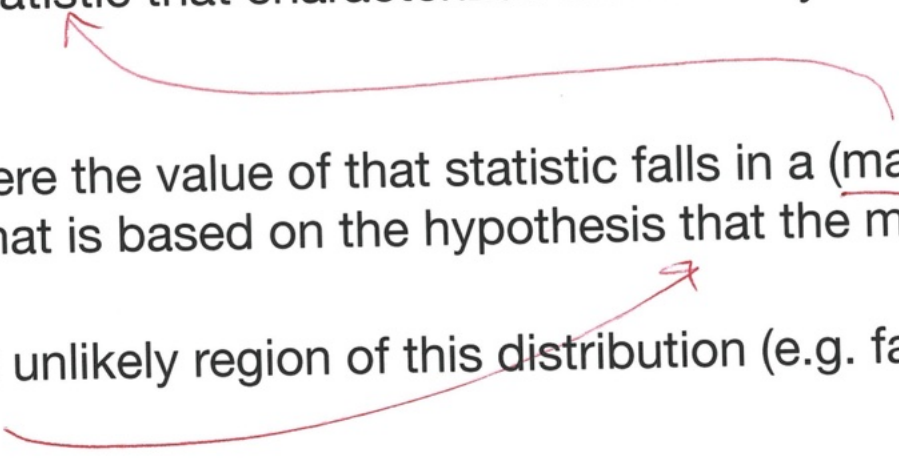
$$Q \equiv 1 - P(x, k) = P(\chi^2 > x)$$

probability of finding a χ^2 at least as bad as this one by chance (bad luck)

If $Q \gtrsim 0.1$: ☺ (model consistent with data)

If $Q \lesssim 0.01$: ☹ (could be inconsistent, or bad luck!)

General goodness of fit

- Same general procedure:
 1. Compute a statistic that characterizes the fit of my model to my data (e.g. a likelihood).
 2. Compute where the value of that statistic falls in a (matching!) probability distribution that is based on the hypothesis that the model is correct.
 3. If it falls in an unlikely region of this distribution (e.g. far in the tail), reject this hypothesis.
 - Is the data **consistent** with being drawn from my model?
- 

Comparing a model with binned data

- A choice of statistic

Any statistic for goodness of fit is fair game, but different statistics may give different answers!

- (Pearson's) chi-square statistic:

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

observed # in bin i (data, integer!!)
expected count from model (can be continuous)
 $E_i \leftarrow$ Poisson! so $\sigma_i^2 = \mu_i = E_i$

- Matching distribution?

χ^2 distr.? No, $\cap$

Only asymptotically converges to χ^2 distr. if $E_i \rightarrow \infty$ for all i
(Actual distr. for Pearson's χ^2 is unknown / has no analytical form)

Deriving distributions for statistics is very hard $\Rightarrow$ works better to just choose the χ^2 distribution, and find a statistic that matches it!

Better convergence to the χ^2 distribution

$$Y^2 = k + \sqrt{\frac{2k}{2k + \sum_i E_i^{-1}}} (\overset{\text{Pearson's}}{\chi^2} - k)$$

Using $x = Y^2$ in χ^2 distr. converges faster than Pearson's χ^2 does (but still not fast)

Recommend: G-test: $G = -2 \sum_i O_i \ln\left(\frac{E_i}{O_i}\right) = 2 \sum_i O_i \ln\left(\frac{O_i}{E_i}\right)$

int !!

Use $x = G$ in χ^2 distr. $\Rightarrow$ calculate Q

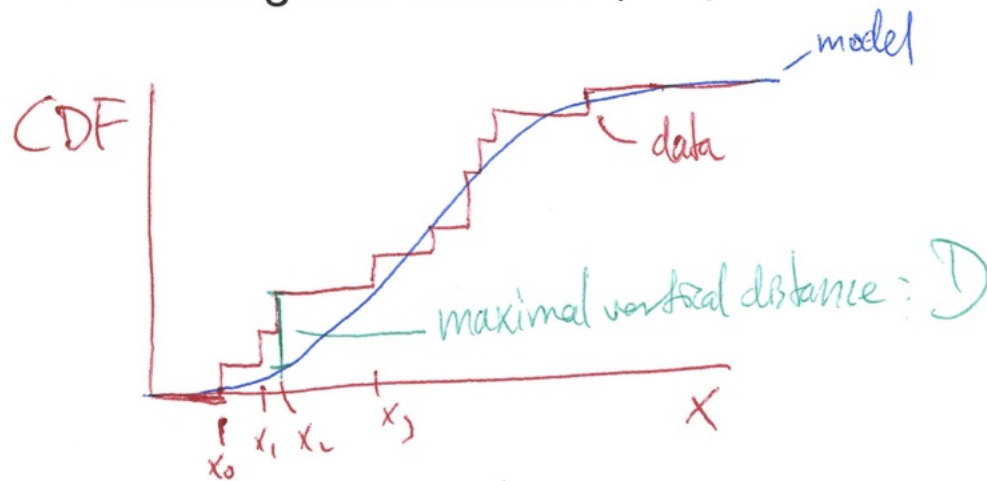
to get model fit significance (=goodness of fit)

G converges much faster than Pearson's χ^2 or Y^2 !

(faster converging statistics exist, but are more complicated)

Comparing a model with unbinned data

- Kolmogorov-Smirnov (K-S) test



K-S statistic: $D = \max$ abs. vertical distance
between data CDF and model CDF
(only need to check for D at the x_i !)

- Useful properties

- (1) invariant for transformations of x (e.g. $x \rightarrow \log x$)
- (2) distribution of D is known to very good approx.!

But: K-S test not
very sensitive to differences
in the tails (CDF is
quite flat there)

K-S distribution

- Cumulative distribution function:

$$P_{\text{KS}}(z) = \frac{\sqrt{2\pi}}{z} \sum_{j=1}^{\infty} \exp\left(-\frac{(2j-1)^2\pi^2}{8z^2}\right) > \text{equiv.}$$
$$= 1 - 2 \sum_{j=1}^{\infty} (-1)^{j-1} \exp(-2j^2z^2)$$

infinite sums!

- Numerically optimal:

K-S distribution

great approx. for $N \geq 4$ already!

$$Z \approx [-\sqrt{N} + 0.12 + 0.11/\sqrt{N}] D$$

- Cumulative distribution function:

$$P_{KS}(z) = \frac{\sqrt{2\pi}}{z} \sum_{j=1}^{\infty} \exp\left(-\frac{(2j-1)^2 \pi^2}{8z^2}\right)$$

$$= 1 - 2 \sum_{j=1}^{\infty} (-1)^{j-1} \exp(-2j^2 z^2)$$

- Numerically optimal:

$$P_{KS}(z) \approx \begin{cases} \frac{\sqrt{2\pi}}{z} \left[\underbrace{\left(e^{-\pi^2/(8z^2)} \right)}_{\text{calculate only once per } z!} + \left(e^{-\pi^2/(8z^2)} \right)^9 + \left(e^{-\pi^2/(8z^2)} \right)^{25} \right] & (z < 1.18) \\ 1 - 2 \left[\underbrace{\left(e^{-2z^2} \right)}_{\text{calculate only once per } z!} - \left(e^{-2z^2} \right)^4 + \left(e^{-2z^2} \right)^9 \right] & (z \geq 1.18) \end{cases}$$

trunc. error $\propto \frac{1}{z^6}$ for doubles!

- Significance

$$Q = 1 - P_{KS}(z)$$

all statistical tests so far were ~ frequentist

The Bayesian approach

- Sample the Bayesian posterior → e.g. through MCMC

PPP: Posterior Predictive p-values (replace frequentist p-values)

- Test quantity vs test statistic

└ some $T(y_0, \underline{\theta}_0)$
original dataset —┐ └ original best-fit params.

For every set $\underline{\theta}_{(i)}$, generate a new data set $y_{(i)}$, and compare $T(y_{(i)}, \underline{\theta}_{(i)})$ to $T(y_0, \underline{\theta}_0)$

**Next week:
question hour!**

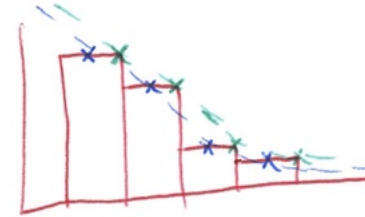
Don't forget : register for every exam ≥ 10 days before!

Bonus: On the placement of data bins

↳ not exam material – but useful

- What value x_i is representable for bin i ?

The linear bin center? The logarithmic bin center? A bin edge?



- Bin value for infinite precision:

$$\bar{y} = \frac{\int_{x_1}^{x_2} f(x) dx}{\int_{x_1}^{x_2} dx} = \frac{\int_{x_1}^{x_2} f(x) dx}{x_2 - x_1} \quad (= \text{weighted mean over the bin})$$

- Example: polynomial

$$\bar{y} = \sum_{i=0}^n \frac{c_i}{i+1} \frac{x_2^{i+1} - x_1^{i+1}}{x_2 - x_1} = \sum_{i=0}^n \frac{c_i}{i+1} \sum_{j=0}^i x_1^j x_2^{i-j}$$

To fit this polynomial 100% accurately (bias-free), we need an $\bar{x}$ for each bin that matches its $\bar{y}$

Candidates

- Mean, log-mean, and weighted average: *(some example candidates for $\bar{x}$)*

$$\bar{x}_{\text{lin}} = \frac{1}{2}(x_1 + x_2) \quad \bar{x}_{\text{log}} = \sqrt{x_1 x_2}$$

$$\tilde{x} = \frac{\int_{x_1}^{x_2} f(x) x \, dx}{\int_{x_1}^{x_2} f(x) \, dx} \quad \left(= \frac{\sum_{i=0}^n \frac{c_i}{i+2} (x_2^{i+2} - x_1^{i+2})}{\sum_{i=0}^n \frac{c_i}{i+1} (x_2^{i+1} - x_1^{i+1})} \right)$$

$\Rightarrow$ none exactly match $\bar{y}$ in general (i.e. none always give $f(\bar{x}) = \bar{y}$)

- Example: polynomial

$$f(\bar{x}) = \sum_{i=0}^n \frac{c_i}{2^i} \sum_{j=0}^i \binom{i}{j} x_1^j x_2^{i-j} \quad \leftarrow \text{only equal to } \bar{y} \text{ if } n=1 \text{ (straight line)}$$

- Use the right value for the function!

Note: for power laws, $\bar{x}_{\text{log}}$ is close to optimal (exact for $n=2$)

Best: calculate $\bar{y}$ for your function, and invert $f(\bar{x}) = \bar{y}$ for $\bar{x}$.